Process FMEA - Bracket Welding Cell - Section 1: Executive Summary & Scope

Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: Highest RPN failure mode is inconsistent weld penetration from electrode wear. Secondary control 1.2: Mitigation adds an automated electrode wear check every 500 cycles.

- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Mitigation adds an automated electrode wear check every 500 cycles.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Process FMEA - Bracket Weldi Gate 1	Carlos Ibarra	3 business days	Active
Process FMEA - Bracket Weldi Gate 2	Carlos Ibarra	10 business days	Monitored

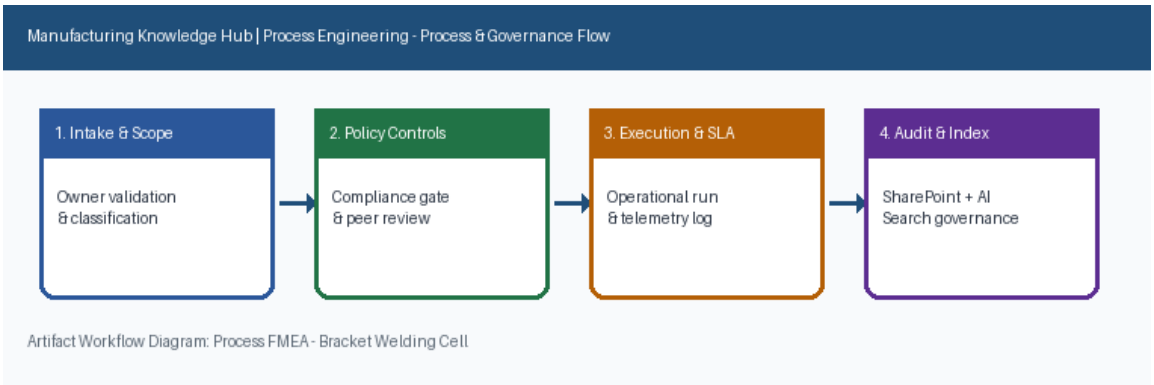
Process FMEA - Bracket Welding Cell - Section 2: Architecture & Process Flow

Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Mitigation adds an automated electrode wear check every 500 cycles. Secondary control 2.2: Detection controls were upgraded from manual visual check to inline vision inspection.

- Mitigation adds an automated electrode wear check every 500 cycles.
- Detection controls were upgraded from manual visual check to inline vision inspection.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Process FMEA - Bracket Weldi Gate 1	Carlos Ibarra	3 business days	Active
Process FMEA - Bracket Weldi Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-2	Carlos Ibarra	3 business days	Stage 2 Verified



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Process FMEA - Bracket Welding Cell - Section 3: Governance Standards & Controls

Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Detection controls were upgraded from manual visual check to inline vision inspection. Secondary control 3.2: FMEA is reviewed and updated after any process or equipment change.

- Detection controls were upgraded from manual visual check to inline vision inspection.
- FMEA is reviewed and updated after any process or equipment change.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Process FMEA - Bracket Weldi Gate 1	Carlos Ibarra	3 business days	Active
Process FMEA - Bracket Weldi Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-3	Carlos Ibarra	10 business days	Stage 3 Verified

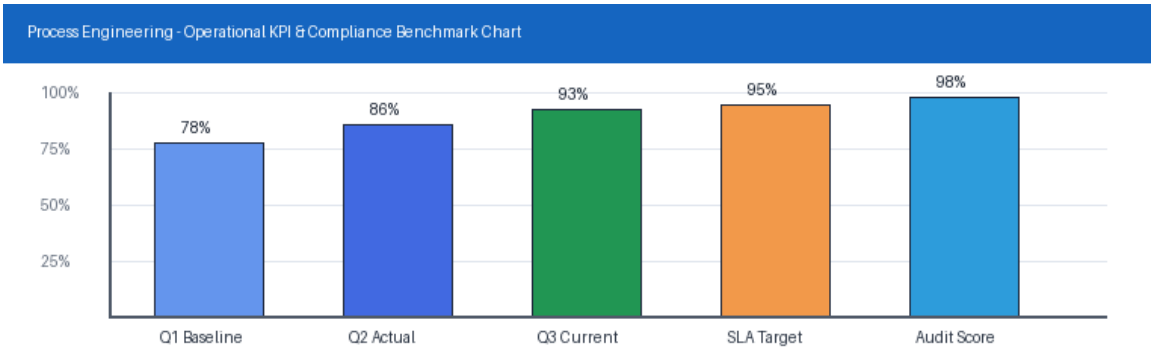
Process FMEA - Bracket Welding Cell - Section 4: Appendix, Lineage & Sign-off

Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 4 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 4.1: FMEA is reviewed and updated after any process or equipment change. Secondary control 4.2: Highest RPN failure mode is inconsistent weld penetration from electrode wear.

- FMEA is reviewed and updated after any process or equipment change.
- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Process FMEA - Bracket Weldi Gate 1	Carlos Ibarra	3 business days	Active
Process FMEA - Bracket Weldi Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-4	Carlos Ibarra	5 business days	Stage 4 Verified



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